

Next-best-view methods occupy a multi-axis design space

The axes describe what is optimized, where the target comes from, how actions are supported, and what state is represented.

Utility

objective

coverage / visibility

uncertainty / information

scene-wide reconstruction

target-specific quality

Target source

conditioning signal

scene-wide objective

oracle-defined target

actor-observed target

target + identity context

Action / horizon

decision support

single greedy view

finite candidate set

bounded lookahead

continuous / hierarchical
control

Representation

state evidence

point / surface

occupancy / ray grid

radiance field / 3DGS

egocentric foundation
evidence

ARIA-NBV position

finite candidates · target-quality utility · actor/oracle-separated target source · egocentric state